

Dmytro Zhuk

ARTIFICIAL INTELLIGENCE RESEARCHER · APPLIED DATA SCIENTIST

Vienna, Austria

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Summary

Greetings! My name is Dmytro Zhuk, and I am currently pursuing a Master's degree in Data Science at the University of Vienna. I graduated from Kyiv National Economic University with a Bachelor's degree in System Analysis with distinction.

My research interests include but are not limited to: Human-Computer Interaction, Explainable and Transparent Artificial Intelligence, Natural Language Processing, Stochastic Processes, and Reinforcement Learning. I am interested in understanding and building systems that can be trusted in complex, high-stakes settings, using computational and data-driven approaches to address problems at the intersection of science, technology, and society.

Communicative, responsible, problem-solving, teamwork, creative, have great attention to details, able to work both in team and independently. I have analytical thinking, persistent in achieving goals, and optimist myself. Also, I like jazz music, opera, Italian art, and travelling.

Education

University of Vienna

Vienna, Austria

MASTER'S DEGREE IN DATA SCIENCE

Jan. 2025 - 2027

- Attending Master's program in Data Science.

Kyiv National Economic University

Kyiv, Ukraine

BACHELOR'S DEGREE IN SYSTEM ANALYSIS

Aug. 2020 - Jun. 2024

- Completed undergraduate studies in System Analysis.

Experience

University of Cambridge, Nokia Bell Labs

Cambridge, United Kingdom

VISITING RESEARCHER

Jul. 2026 - Current

- Researching value visualization for shaping user perceptions of large language models.
- Results to be presented at an international conference (2 peer-reviewed publications).

Imperial College London

London, United Kingdom

VISITING RESEARCHER

Mar. 2026 - Current

- Developed a malware labeling framework using threat intelligence and uncertainty-aware reasoning.
- Analyzed 130k VirusTotal reports to estimate labeling noise.
- Results to be submitted to an international journal (1 peer-reviewed publication).

CISPA - Helmholtz Center for Information Security

Saarbrücken, Germany

VISITING WINTER RESEARCHER

Nov. 2025 - Feb. 2026

- Fine-tuned open-source large language models on GLUE benchmark tasks for dataset inference.
- Prepared a baseline inference task for the European Championship in Trustworthy AI, evaluated using TPR@FPR=1%.

CERN - European Organization for Nuclear Research

Geneva, Switzerland

VISITING SUMMER RESEARCHER

Jul. 2025 - Oct. 2025

- Optimized full-text retrieval for 500+ scientific papers, reducing systematic review time by up to 80%.
- Developed intelligent components to automate systematic review workflows.
- Presented results at an international conference (1 peer-reviewed publication).

SBA Research - Austria's largest dedicated cybersecurity research center

Vienna, Austria

R&D CYBERSECURITY ANALYST

Mar. 2025 - Current

- Designed domain scoring models for 30k+ domains/day with over 80% recall.
- Applied LLMs for automated threat reporting and analyzed 100+ DGA campaigns.

Whalebone - award-winning cybersecurity company providing DNS security

Brno, Czechia

THREAT INTELLIGENCE SPECIALIST

Dec. 2022 - Sep. 2024

- Built a domain classification system achieving over 84% accuracy.
- Developed clustering, time-series, and threat intelligence solutions for domain monitoring.

Research Schools

2026 **Pre-doctoral Research School in Computer Science**, Cornell, Maryland, Max Planck

2026 **Advanced School in Uncertainty of DL Models for Science**, International Centre for Theoretical Physics

Skills

Programming Languages	Python, Java, Matlab
Machine Learning & Deep Learning	Scikit-Learn, Tensorflow, Keras, Pytorch
Natural Language Processing & LLMs	NLTK, spaCy, Gensim, Transformers HF, Ollama, Langchain, Chroma
Data Mining	Beautiful Soup, Urllib3, Selenium
Data Analysis	Pandas, Numpy, Polars
Data Visualization	Tableau, Microsoft PowerBI, Kibana, Plotly, Seaborn, Gradio
Software Development	Git, Linux, Docker, Modal, K8s, Rancher
Databases	SQL, NoSQL, Elasticsearch, MySQL, Microsoft SQL Server
Languages	English (Fluent), Czech (Fluent), German (Good), Ukrainian (Native), Russian (Native)
Certificates	LLM Engineering, ML Engineer Career Track, High-Dimensional Data Analysis, AGI Strategy

Journal Articles

Temporal Dynamics-Based Noise Estimation in Large-Scale Malware APK Datasets	2027 (TBD)
DMYTRO ZHUK, FAHAD ALOTAIBI, AND SEGIO MAFFEIS	
• IEEE Security & Privacy Journal.	
Random Processes in Meteorology	2022
VOLODYMYR HALITSYN, DMYTRO ZHUK, AND ANNA PETRYCHENKO	
• Modeling and Information System in Economics Journal.	

Conference Proceedings

From Displayed Values to Behavioral Alignment: Dynamic Value Visualisation for Hybrid Human-AI Systems	Europe
DMYTRO ZHUK, EDYTA PAULINA BOGUCA, AND DR MALAK SADEK	
• Conference on Neural Information Processing Systems (NeurIPS).	
Same Chatbot, Different Experience: How Visualising LLM Chatbot Values Shapes User Perceptions	Pittsburgh, United States
QINGFENG HUANG, DMYTRO ZHUK, EDYTA PAULINA BOGUCA, JATINDER SINGH, AND DR MALAK SADEK	
• ACM Conference on Human Factors in Computing Systems (ACM CHI).	
Automating License-Aware Full-Text Retrieval for Systematic Reviews: An End-To-End Scalable System to Reduce Reviewer Workload	Helsinki, Finland
DMYTRO ZHUK, ELIAS SANDNER, IGOR JAKOVljeVIC, ALICE SIMNICEANU, LUCA FONTANA, ANDRE HENRIQUES, ANDREAS WAGNER, AND CHRISTIAN GÜTL	
• International Open Search Symposium (OSSYM).	
Model for Food Supply Chain Stability Under Climate Variability with Semi-Markov Processes	Kyiv, Ukraine
MIROSLAVA RŮŽIČKOVÁ, IRADA DZHALLADOVA, AND DMYTRO ZHUK	
• Dynamical System Modeling and Stability Investigation International Conference (DSMSI).	
Designing Trustworthy Recommender System for Smart City Sustainability	Lviv, Ukraine
DMYTRO ZHUK AND IRADA DZHALLADOVA	
• Information and Measurement Technologies International Conference (LPNU).	
The Ethics of Artificial Intelligence: Challenges and Frameworks	Kyiv, Ukraine
DMYTRO ZHUK	
• Artificial Intelligence and Intelligence Systems International Conference (AIIS).	

Honors & Awards

2026	Full Scholarship , EAGxNordics AI Safety
2024	Honours Bachelor's Degree, Best Bachelor's Thesis Awards , Kyiv National Economic University
2021	Best Presentation and Poster Award, 1st Place , Intelligent Methods of Analysis of Cybersecurity Threats Conference
2020	Fully Funded Scholarship , Kyiv National Economic University